

# AWS EC2 Hosted Campaign Registration Portal with Apache, PHP, IAM Role, and S3 Logging

---

## Technologies Used

| Component      | Technology          |
|----------------|---------------------|
| Cloud Platform | Amazon Web Services |
| Server         | Amazon EC2          |
| Storage        | Amazon S3           |
| Web Server     | Apache2             |
| Backend        | PHP                 |
| Frontend       | HTML + CSS          |
| Authentication | IAM Role            |
| OS             | Ubuntu 22.04        |

---

## PART 1 — CREATE S3 BUCKET

### Step 1 — Open S3 Console

Navigate:

AWS Console → S3

Click:

Create bucket

---

## Step 2 — Configure Bucket

Bucket Name:

capstone-project-milestone-v

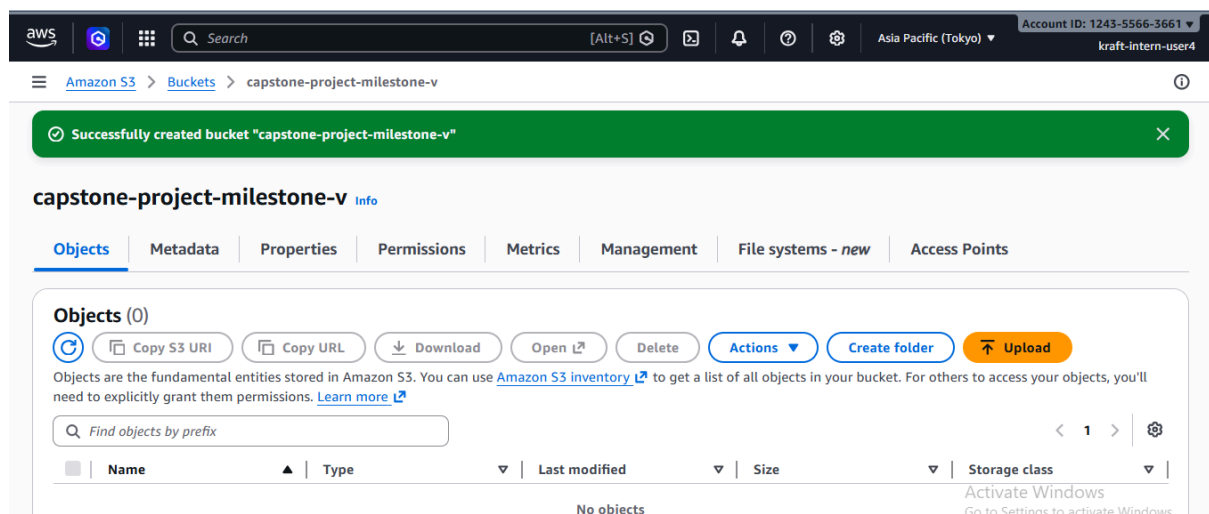
Keep everything default.

Click:

Create bucket

---

## Expected Output



S3 bucket created successfully.

---

## PART 2 — CREATE EC2 INSTANCE

## Step 3 — Open EC2

Navigate:

AWS Console → EC2

Click:

Launch Instance

---

## Step 4 — Configure EC2

| Setting          | Value                  |
|------------------|------------------------|
| Name             | campaign-server-v      |
| AMI              | Ubuntu Server<br>22.04 |
| Instance<br>Type | t2.micro               |

---

## Step 5 — Create Key Pair

Click:

Create new key pair

Name:

campaign-key-v

Download PEM file.

---

## Step 6 — Configure Security Group

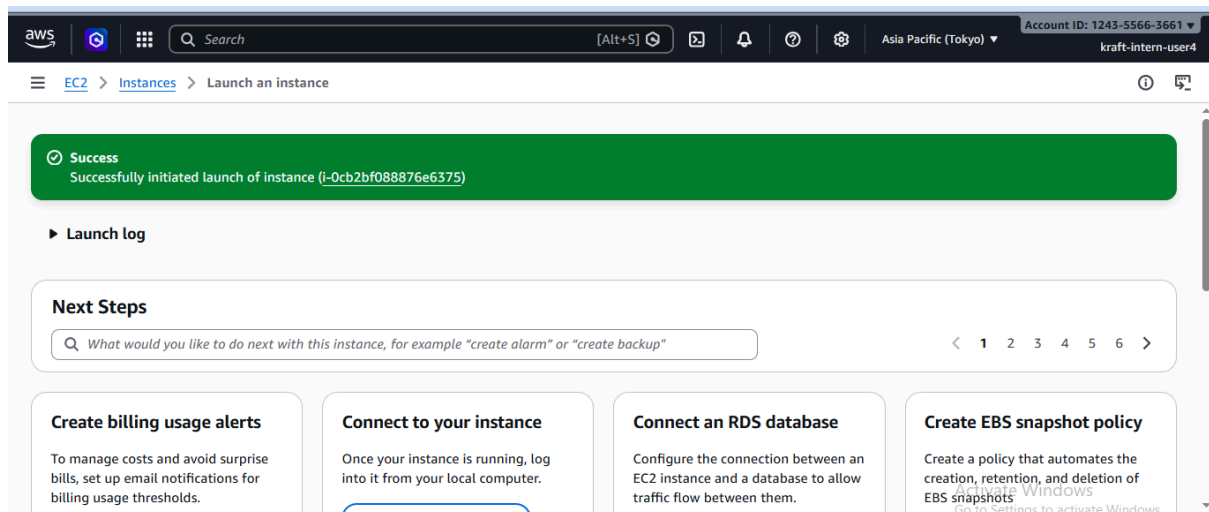
Allow:

| Type | Port |
|------|------|
| SSH  | 22   |
| HTTP | 80   |

Launch instance.

---

## Expected Output



EC2 instance running successfully.

---

## PART 3 — ATTACH IAM ROLE TO EC2

### Step 7 — Attach Role

Navigate:

EC2 → Instances

Select instance.

Click:

Actions → Security → Modify IAM Role

Attach:

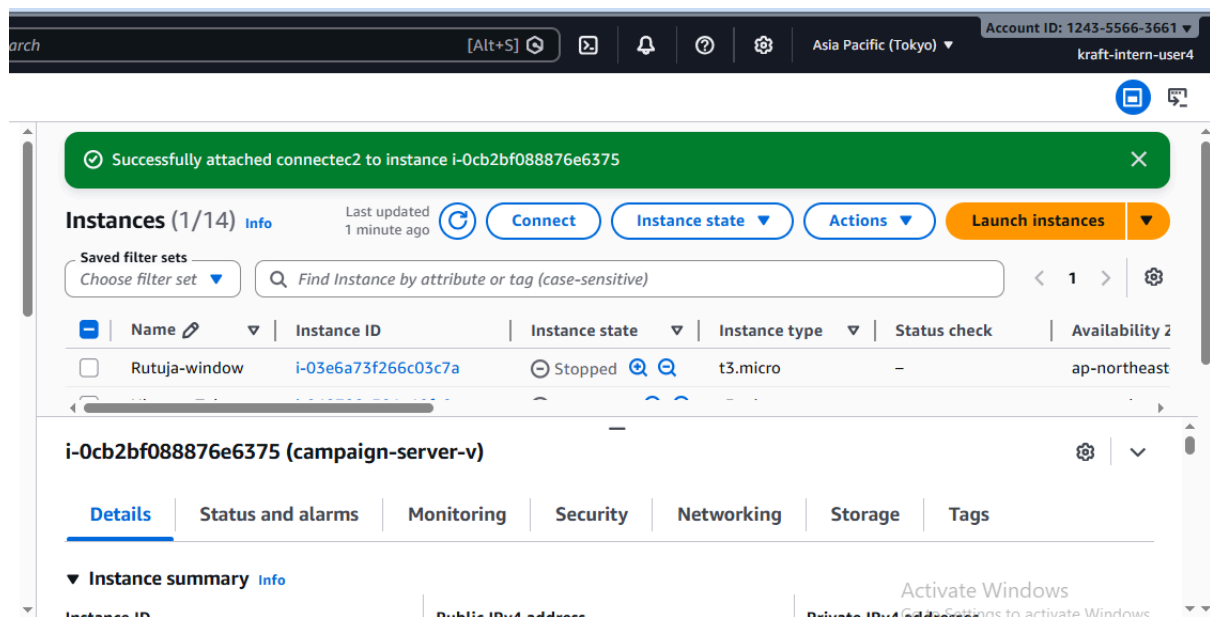
connectec2

Click:

Update IAM Role

---

## Expected Output



IAM Role attached successfully.

---

## PART 4 — CONNECT TO EC2

## Step 8 — Open Terminal

Move to Downloads:

```
cd Downloads
```

---

## Step 9 — Set PEM Permissions

Run:

```
chmod 400 campaign-key.pem
```

---

## Step 10 — Connect to EC2

Run:

```
ssh -i campaign-key.pem ubuntu@<EC2-PUBLIC-IP>
```

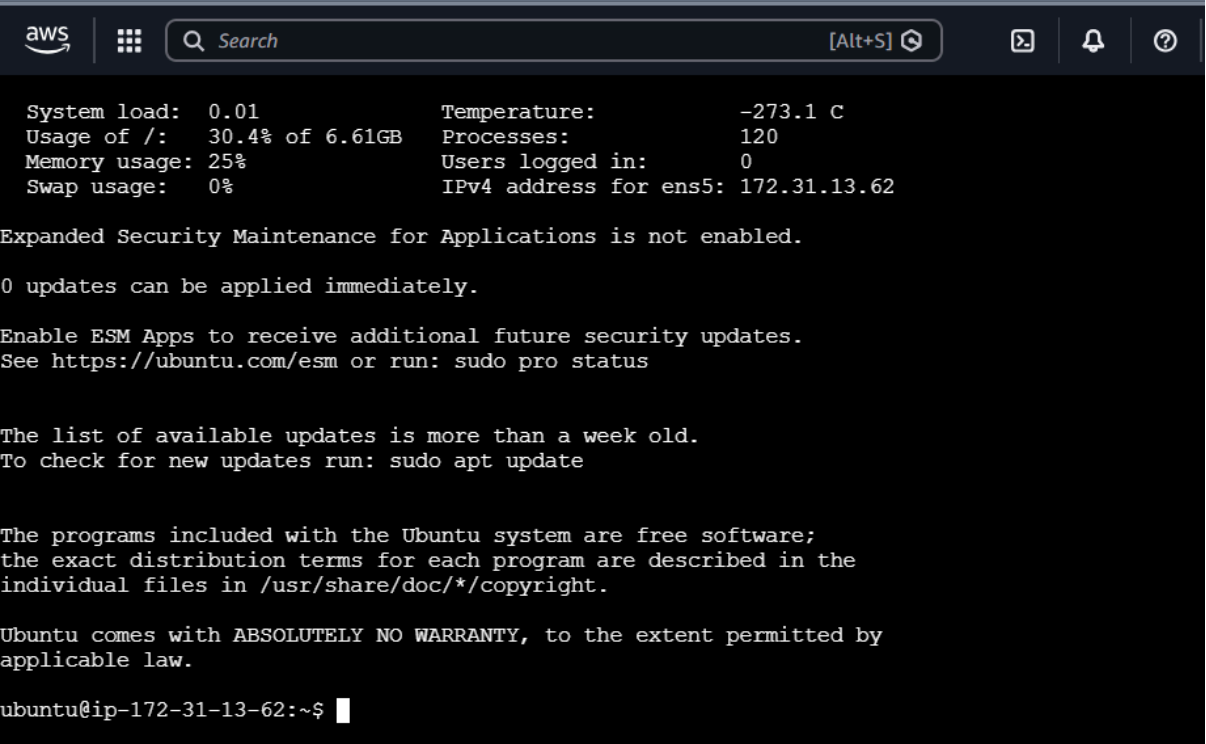
Example:

```
ssh -i campaign-key.pem ubuntu@35.77.24.74
```

```
ssh -i campaign-key-v.pem ec2-user@18.183.112.9
```

---

## Expected Output



The screenshot shows an AWS CloudShell terminal window. The top bar includes the AWS logo, a search bar, and navigation icons. The terminal output displays system statistics: System load (0.01), Usage of / (30.4% of 6.61GB), Memory usage (25%), Swap usage (0%), Temperature (-273.1 C), Processes (120), Users logged in (0), and IPv4 address for ens5 (172.31.13.62). It also shows that Expanded Security Maintenance for Applications is not enabled, 0 updates can be applied immediately, and provides instructions to enable ESM Apps and check for updates. A disclaimer about Ubuntu's warranty is also present. The prompt is ubuntu@ip-172-31-13-62:~\$.

```
aws | [Search] | [Alt+S] | [Icons]

System load: 0.01      Temperature: -273.1 C
Usage of /: 30.4% of 6.61GB Processes: 120
Memory usage: 25%     Users logged in: 0
Swap usage: 0%        IPv4 address for ens5: 172.31.13.62

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-172-31-13-62:~$
```

Ubuntu terminal opens successfully.

---

## PART 5 — INSTALL SOFTWARE

### Step 11 — Update Packages

Run:

```
sudo apt update -y
```

---

### Step 12 — Install Apache + PHP + AWS CLI

Run:

```
sudo apt install apache2 php awscli -y
```

---

## Step 13 — Start Apache

Run:

```
sudo systemctl start apache2
```

Enable Apache:

```
sudo systemctl enable apache2
```

---

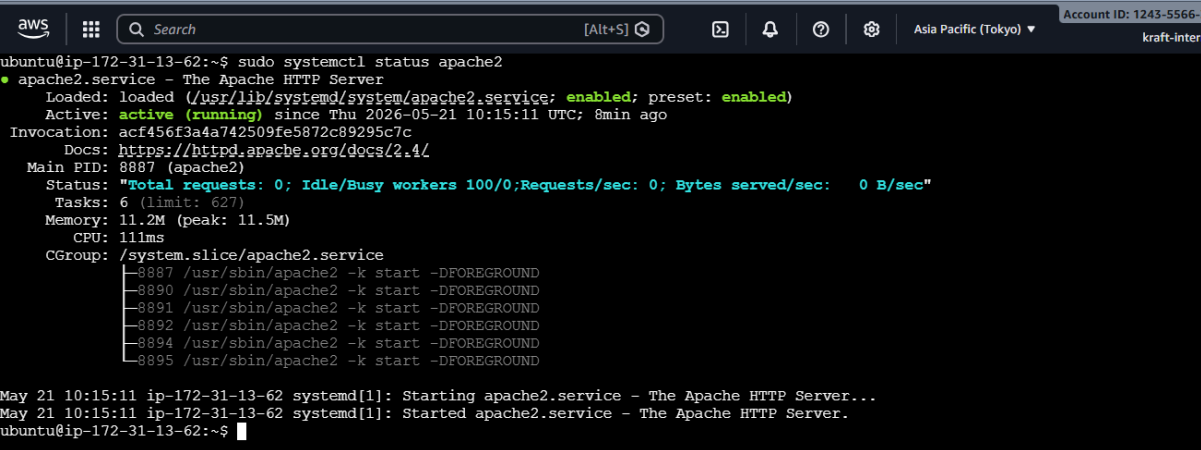
## Step 14 — Check Apache Status

Run:

```
sudo systemctl status apache2
```

Expected:

active (running)



```
aws | Search [Alt+S] | Asia Pacific (Tokyo) | Account ID: 1243-5566-3 | kraft-interr
ubuntu@ip-172-31-13-62:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-05-21 10:15:11 UTC; 8min ago
   Invocation: acf456f3a4a742509fe5872c89295c7c
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 8887 (apache2)
   Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec: 0 B/sec"
   Tasks: 6 (limit: 627)
  Memory: 11.2M (peak: 11.5M)
    CPU: 111ms
   CGroup: /system.slice/apache2.service
           └─8887 /usr/sbin/apache2 -k start -DFOREGROUND
             └─8890 /usr/sbin/apache2 -k start -DFOREGROUND
               └─8891 /usr/sbin/apache2 -k start -DFOREGROUND
                 └─8892 /usr/sbin/apache2 -k start -DFOREGROUND
                   └─8894 /usr/sbin/apache2 -k start -DFOREGROUND
                     └─8895 /usr/sbin/apache2 -k start -DFOREGROUND

May 21 10:15:11 ip-172-31-13-62 systemd[1]: Starting apache2.service - The Apache HTTP Server...
May 21 10:15:11 ip-172-31-13-62 systemd[1]: Started apache2.service - The Apache HTTP Server.
ubuntu@ip-172-31-13-62:~$
```

i-0cb2bf088876e6375 (campaign-server-v)

PublicIPs: 18.183.112.9 PrivateIPs: 172.31.13.62

Activate Windows  
Go to Settings to activate Windows

Press:

Q



to exit.

---

## PART 6 — VERIFY WEBSITE

### Step 15 — Open Browser

Open:

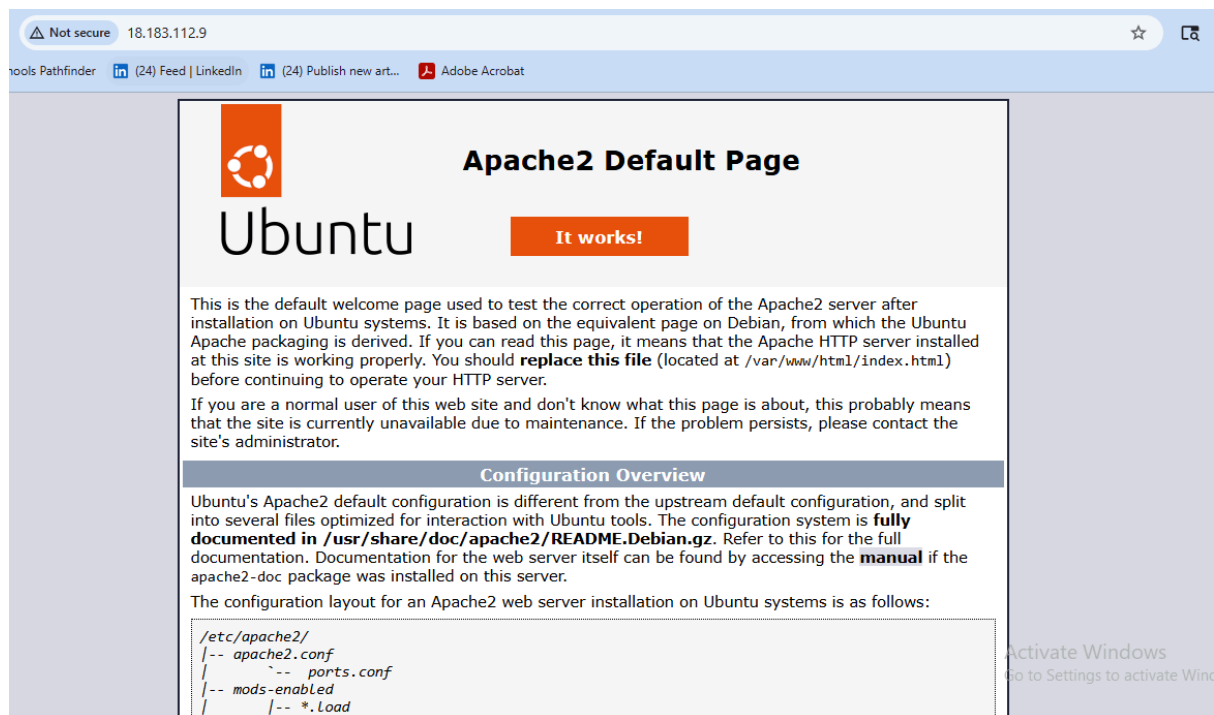
http://<EC2-PUBLIC-IP>

Example:

http://35.77.24.74

---

### Expected Output



Apache default page appears.

---

## PART 7 — CREATE WEBSITE FILES

### Step 16 — Move to Apache Directory

Run:

```
cd /var/www/html
```

---

### Step 17 — Remove Default Apache Page

Run:

```
sudo rm index.html
```

---

## PART 8 — CREATE HTML PAGE

### Step 18 — Create index.html

Run:

```
sudo nano index.html
```

Paste FULL CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Campaign Registration Portal</title>

  <style>

    *{
```

```
    margin:0;
    padding:0;
    box-sizing:border-box;
}

body{
    font-family: Arial, sans-serif;
    height:100vh;
    display:flex;
    justify-content:center;
    align-items:center;
    background: linear-gradient(135deg,#1e3c72,#2a5298);
}

.container{
    width:420px;
    background:white;
    padding:40px;
    border-radius:20px;
    box-shadow:0 10px 30px rgba(0,0,0,0.3);
}

h1{
    text-align:center;
    color:#1e3c72;
    margin-bottom:10px;
}

p{
    text-align:center;
    color:gray;
    margin-bottom:25px;
}

input{
    width:100%;
    padding:14px;
    margin-top:12px;
    border-radius:10px;
    border:1px solid #ccc;
    font-size:15px;
}

button{
    width:100%;
```

```
    margin-top:20px;
    padding:14px;
    background:linear-gradient(135deg,#1e3c72,#2a5298);
    color:white;
    border:none;
    border-radius:10px;
    font-size:16px;
    cursor:pointer;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
    <h1>Campaign Portal</h1>
```

```
    <p>Register for our latest campaign</p>
```

```
    <form action="submit.php" method="POST">
```

```
        <input type="text" name="name" placeholder="Enter Full Name" required>
```

```
        <input type="email" name="email" placeholder="Enter Email Address" required>
```

```
        <input type="text" name="phone" placeholder="Enter Phone Number" required>
```

```
        <input type="text" name="campaign" placeholder="Enter Campaign Name" required>
```

```
        <button type="submit">
```

```
            Register Now
```

```
        </button>
```

```
    </form>
```

```
</div>
```

```
</body>
```

```
</html>
```

---

## Step 19 — Save File

Press:

CTRL + O

ENTER

CTRL + X

---

## PART 9 — CREATE PHP BACKEND

### Step 20 — Create submit.php

Run:

```
sudo nano submit.php
```

Paste FULL CODE:

```
<?php
```

```
date_default_timezone_set('Asia/Kolkata');
```

```
$name = isset($_POST['name']) ? $_POST['name'] : 'Unknown';
```

```
$email = isset($_POST['email']) ? $_POST['email'] : '';
```

```
$phone = isset($_POST['phone']) ? $_POST['phone'] : '';
```

```
$campaign = isset($_POST['campaign']) ? $_POST['campaign'] : '';
```

```
$data = array(
```

```
    "name" => $name,
```

```
    "email" => $email,
```

```
    "phone" => $phone,
```

```
    "campaign" => $campaign,
```

```
    "timestamp" => date("Y-m-d H:i:s")
```

```
);
```

```
$json_data = json_encode($data, JSON_PRETTY_PRINT);
```

```
$file_name = $name . "_" . time() . ".json";
```

```
$temp_file = "/tmp/" . $file_name;

file_put_contents($temp_file, $json_data);

$bucket_name = "capstone-project-milestone-v";

// FIXED: Added HOME=/tmp to give the AWS CLI a valid runtime working directory
$command = "HOME=/tmp /usr/bin/aws s3 cp \"$temp_file\"
`s3://$bucket_name/logs/$file_name`" > /var/www/html/aws_error.txt 2>&1";

shell_exec($command);

echo "
<!DOCTYPE html>
<html>
<head>
<title>Success</title>
<style>
body{
    font-family:Arial;
    background:linear-gradient(135deg,#1e3c72,#2a5298);
    height:100vh;
    display:flex;
    justify-content:center;
    align-items:center;
}
.box{
    background:white;
    padding:40px;
    border-radius:20px;
    text-align:center;
    width:420px;
}
h1{
    color:green;
    margin-bottom:20px;
}
a{
    display:inline-block;
    margin-top:20px;
    background:#1e3c72;
    color:white;
    padding:12px 20px;
    text-decoration:none;
    border-radius:10px;
```

```
}
</style>
</head>
<body>
<div class='box'>
<h1>Registration Successful</h1>
<p>Campaign data uploaded to S3 successfully.</p>
<p><b>Name:</b> $name</p>
<p><b>Campaign:</b> $campaign</p>
<a href='index.html'>Back to Home</a>
</div>
</body>
</html>
";

?>
```

---

## IMPORTANT

If `/usr/bin/aws` does not work:

Run:

which aws

Example output:

`/usr/local/bin/aws`

Then replace:

`/usr/bin/aws`

with:

`/usr/local/bin/aws`

inside `submit.php`.

---

## Step 21 — Save File

Press:

CTRL + O

ENTER

CTRL + X

---

## PART 10 — VERIFY IAM ACCESS

### Step 22 — Test S3 Access

Run:

aws s3 ls

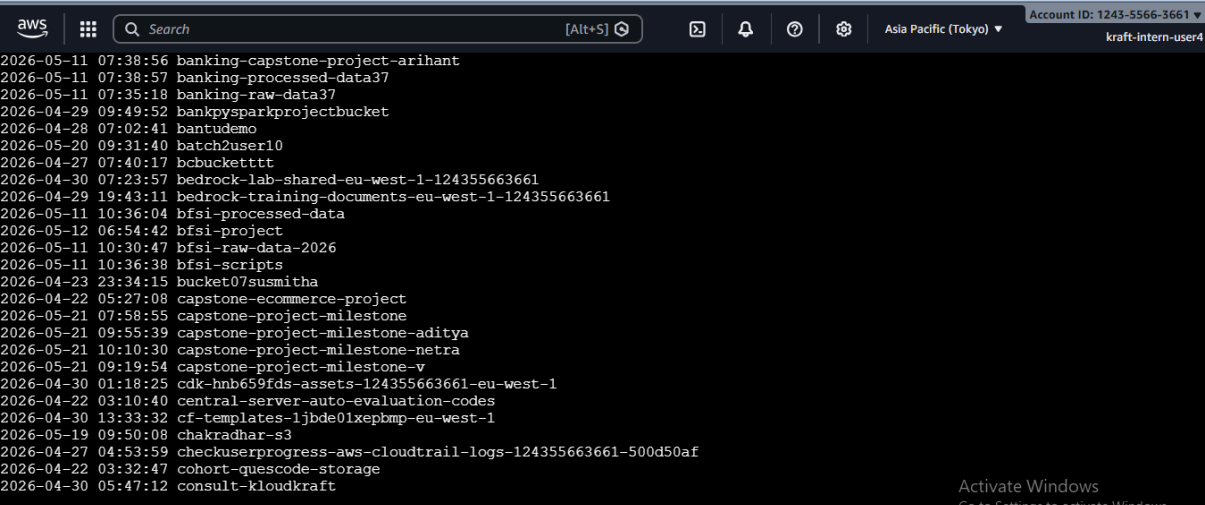
---

## Expected Output

You should see:

Capstone-project-milestone

For me Capstone-project-milestone-v



```
aws | Search [Alt+S] | Asia Pacific (Tokyo) | Account ID: 1243-5566-3661 | kraft-intern-user4
2026-05-11 07:38:56 banking-capstone-project-arihant
2026-05-11 07:38:57 banking-processed-data37
2026-05-11 07:35:18 banking-raw-data37
2026-04-29 09:49:52 bankpysparkprojectbucket
2026-04-28 07:02:41 bantudemo
2026-05-20 09:31:40 batch2user10
2026-04-27 07:40:17 bcbuckettttt
2026-04-30 07:23:57 bedrock-lab-shared-eu-west-1-124355663661
2026-04-29 19:43:11 bedrock-training-documents-eu-west-1-124355663661
2026-05-11 10:36:04 bfsi-processed-data
2026-05-12 06:54:42 bfsi-project
2026-05-11 10:30:47 bfsi-raw-data-2026
2026-05-11 10:36:38 bfsi-scripts
2026-04-23 23:34:15 bucket07susmitha
2026-04-22 05:27:08 capstone-ecommerce-project
2026-05-21 07:58:55 capstone-project-milestone
2026-05-21 09:55:39 capstone-project-milestone-aditya
2026-05-21 10:10:30 capstone-project-milestone-netra
2026-05-21 09:19:54 capstone-project-milestone-v
2026-04-30 01:18:25 cdk-hnb659fds-assets-124355663661-eu-west-1
2026-04-22 03:10:40 central-server-auto-evaluation-codes
2026-04-30 13:33:32 cf-templates-1jbde01xepbmp-eu-west-1
2026-05-19 09:50:08 chakradhar-s3
2026-04-27 04:53:59 checkuserprogress-aws-cloudtrail-logs-124355663661-500d50af
2026-04-22 03:32:47 cohort-quescode-storage
2026-04-30 05:47:12 consult-kloudkraft
Activate Windows
Go to Settings to activate Windows
```



## PART 11 — RESTART APACHE

### Step 23 — Restart Apache

Run:

```
sudo systemctl restart apache2
```

---

## PART 12 — TEST PROJECT

### Step 24 — Open Website

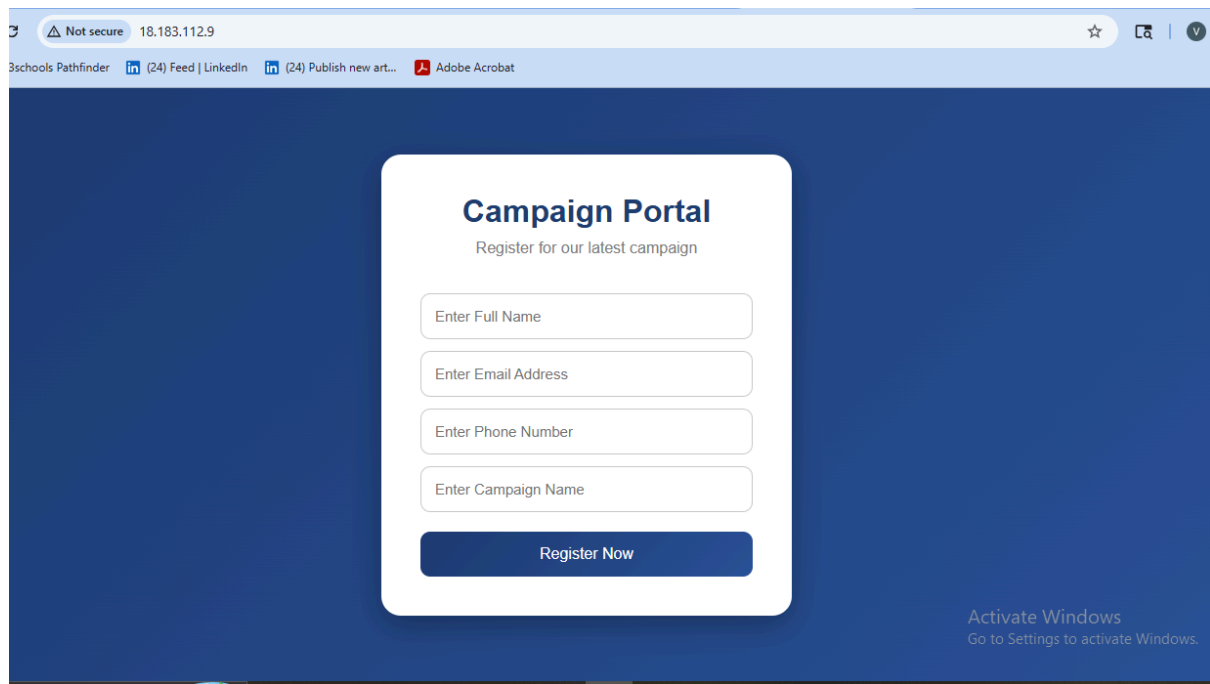
Open:

```
http://35.77.24.74
```

---

### Expected Output

Beautiful registration form appears.



## Step 25 — Submit Form

Enter:

| Field    | Example        |
|----------|----------------|
| Name     | John           |
| Email    | john@gmail.com |
| Phone    | 9999999999     |
| Campaign | Summer Sale    |

Click:

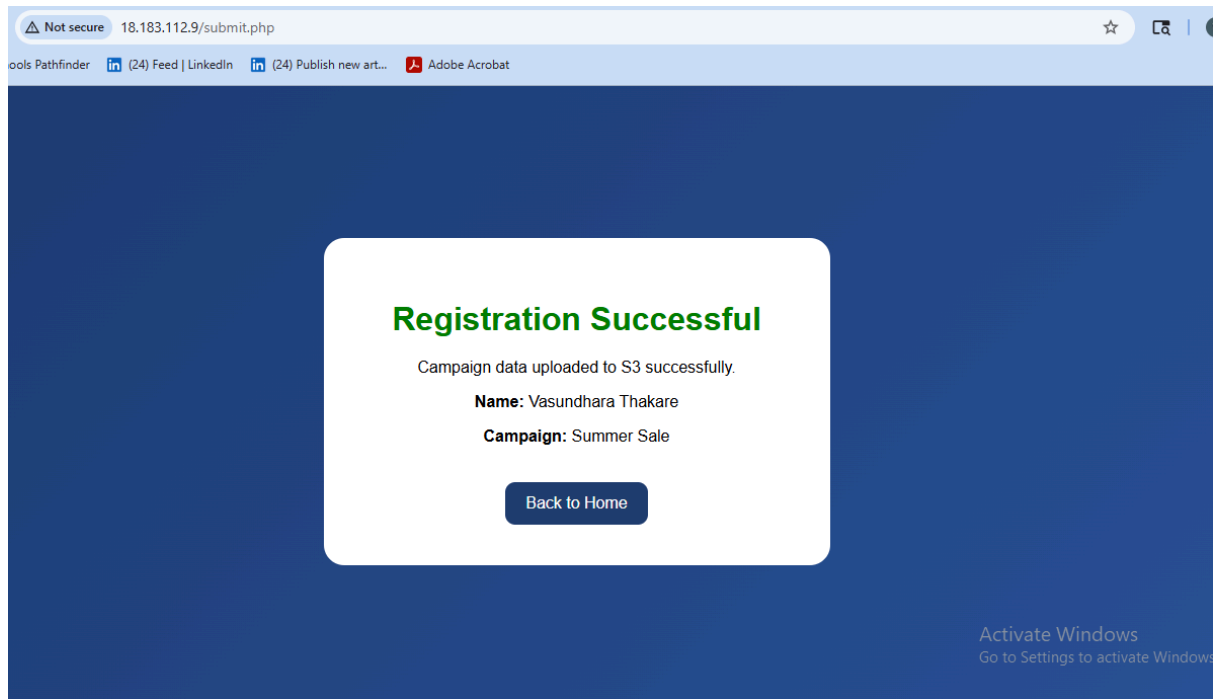
Register Now

## Expected Output

Success page appears:

## Registration Successful

Campaign data uploaded to S3 successfully.



---

## PART 13 — VERIFY LOG FILE

### Step 26 — Open S3 Bucket

Navigate:

S3 → capstone-project-milestone → logs/

You should see:

John\_1779351448.json

---

### Step 27 — Open File

Click JSON file.

Click:

Download

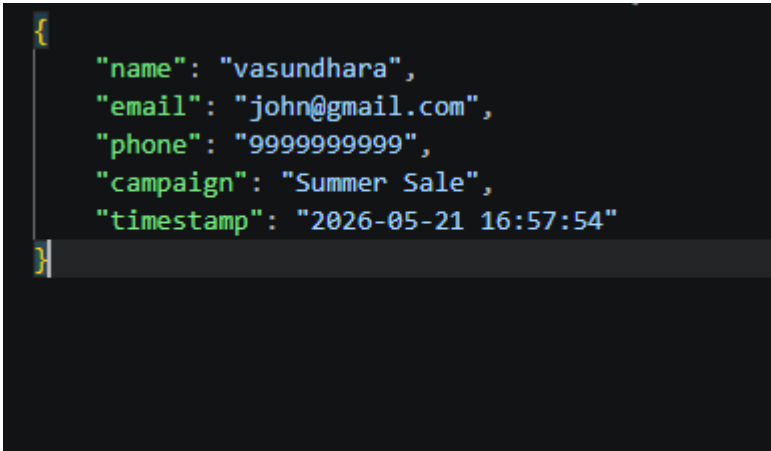
OR

Open

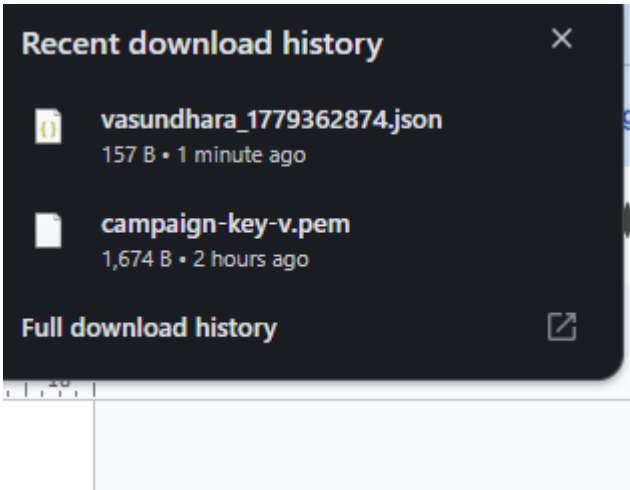
---

## Expected JSON Output

```
{  
  "name": "John",  
  "email": "john@gmail.com",  
  "phone": "9999999999",  
  "campaign": "Summer Sale",  
  "timestamp": "2026-05-21 13:47:30"  
}
```

A screenshot of a code editor with a dark background. It displays a JSON object with the following fields: "name" with value "vasundhara", "email" with value "john@gmail.com", "phone" with value "9999999999", "campaign" with value "Summer Sale", and "timestamp" with value "2026-05-21 16:57:54". The code is color-coded, with opening and closing curly braces in yellow, strings in green, and punctuation in light blue. The editor has a vertical scrollbar on the right side.

```
{  
  "name": "vasundhara",  
  "email": "john@gmail.com",  
  "phone": "9999999999",  
  "campaign": "Summer Sale",  
  "timestamp": "2026-05-21 16:57:54"  
}
```



---

# Final Project Structure

```
/var/www/html/  
|  
├── index.html  
└── submit.php
```

---

# Skills Learned

| Area           | Skills       |
|----------------|--------------|
| AWS            | EC2, S3, IAM |
| Linux          | SSH, Apache  |
| Web Hosting    | Apache2      |
| Frontend       | HTML + CSS   |
| Backend        | PHP          |
| Cloud Storage  | S3           |
| Authentication | IAM Roles    |
| AWS CLI        | S3 Uploads   |

---

